

Three IMAGE futures for IAM-LCA coupling

An eleven-page guide to SSP1-L, SSP2-M and SSP3-H • global results and sector pathways

Scenarios are coherent ‘what-if’ worlds—not predictions. They connect assumptions about society, technology and policy to future energy, material and emissions pathways.

SSP1-L

Sustainability • low warming

Cooperation, resource efficiency and cleaner technologies advance rapidly. Energy and material demand grow more slowly, while mitigation expands across sectors.

TRANSITION PATTERN

Early and coordinated

SSP2-M

Middle road • medium warming

Historical social, economic and technological patterns broadly continue. Development and cleaner technologies advance, but unevenly and without the deep transformation needed for a low-warming future.

TRANSITION PATTERN

Gradual diversification

SSP3-H

Regional Rivalry • high warming

Countries prioritize security and self-sufficiency. International cooperation weakens, development slows and technology diffuses less evenly, making climate action and adaptation more difficult.

TRANSITION PATTERN

Delayed and uneven

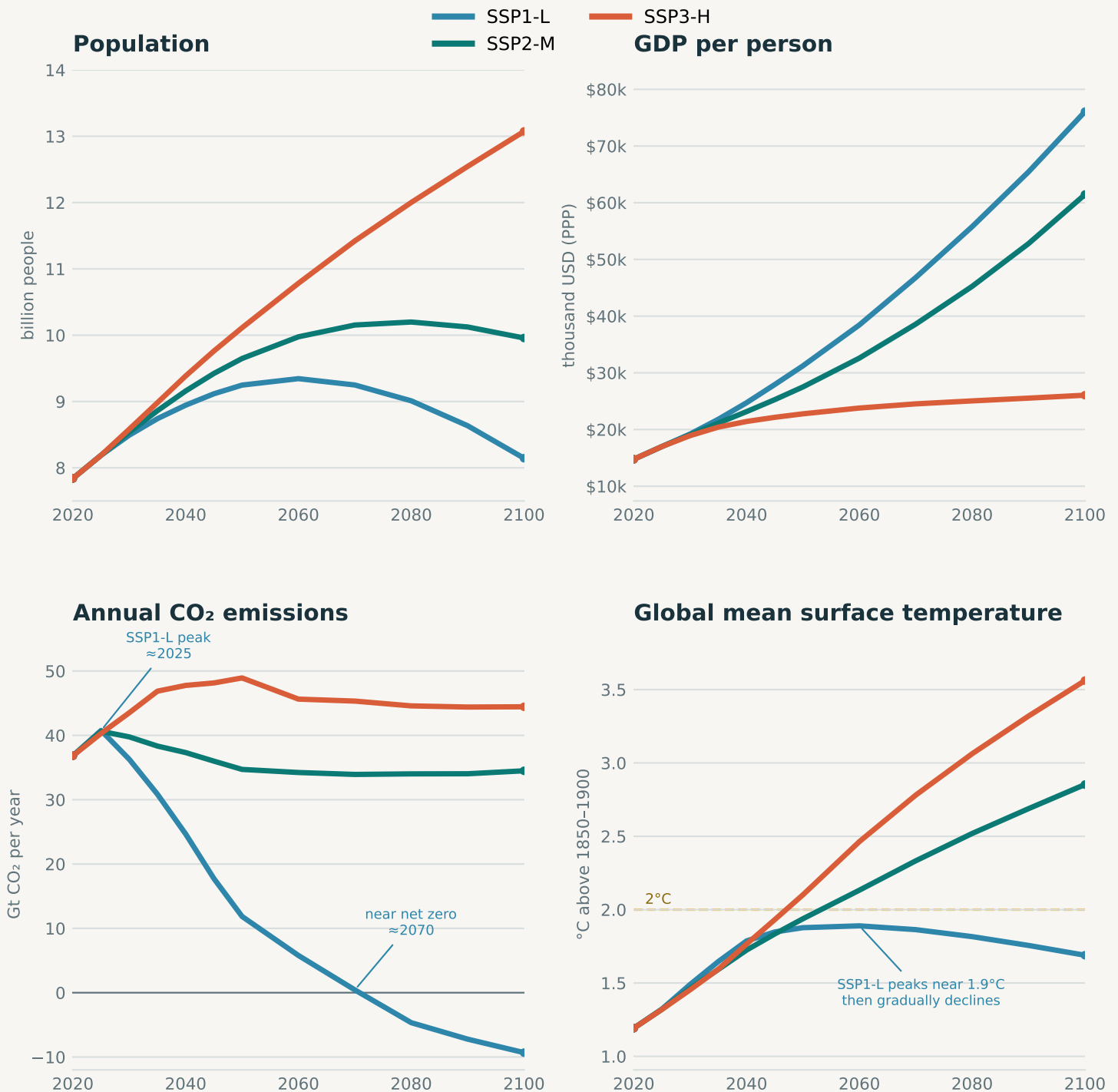
GLOBAL OUTCOMES IN 2100	SSP1-L	SSP2-M	SSP3-H
Population	8.1 bn	10.0 bn	13.1 bn
GDP per person (PPP)	\$76.1k	\$61.4k	\$26.1k
Annual CO ₂	-9.3 Gt	34.5 Gt	44.4 Gt
Warming above 1850–1900	1.69°C	2.85°C	3.56°C

How to read the names

The SSP number describes socioeconomic conditions; L, M and H identify low-, medium- and high-warming pathway families. These are conditional futures, not forecasts. The technology results are IMAGE-specific: another IAM may translate similar assumptions into different demands, technologies and regional patterns.

People, prosperity and climate diverge

The pathways are close in the near term, then separate strongly after mid-century



THE MAIN DIVERGENCE

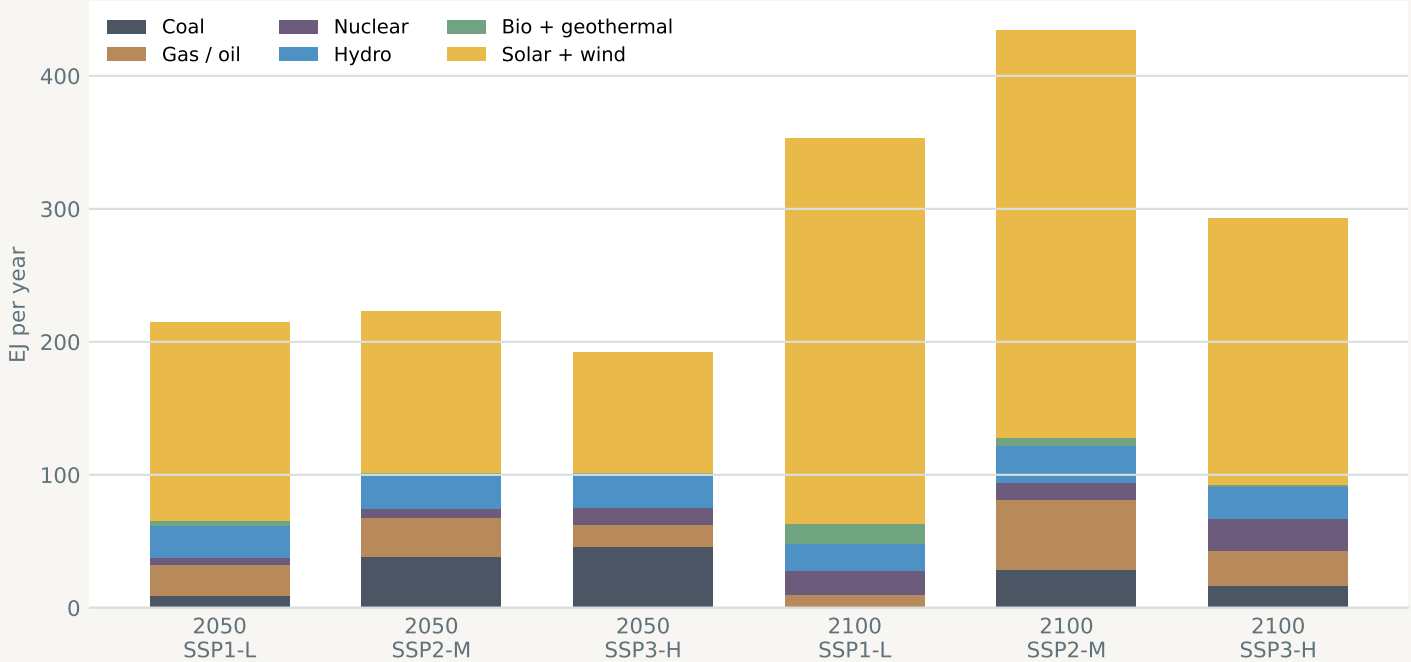
SSP1-L peaks early, approaches net-zero CO₂ around 2070 and becomes net-negative thereafter. Because climate responds with a delay, warming peaks near 1.9°C before declining to 1.69°C by 2100. SSP2-M and SSP3-H remain strongly positive-emissions pathways and reach 2.85°C and 3.56°C, respectively.

Technology pathways reshape future LCA

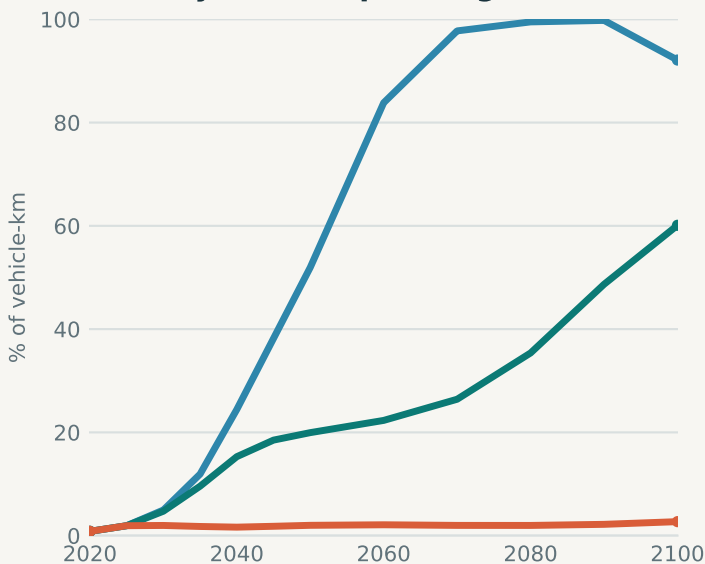
Background inventories respond to deployment, fuel choices and fleet composition—not only total emissions

Global electricity generation by technology

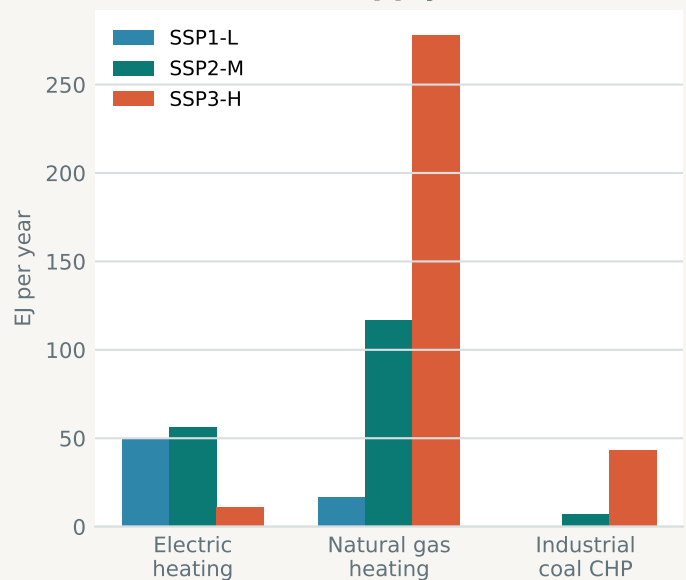
Battery storage excluded to avoid double counting generation



Battery-electric passenger cars



Selected heat supply in 2100



SSP1-L combines rapid renewable deployment with much faster electrification of cars and heat. SSP2-M diversifies more gradually, while SSP3-H retains stronger fossil dependence outside the power sector. Technology composition—not a single global carbon number—determines the background changes inherited by an LCA model.

2050: three materially different backgrounds

A global mid-century scorecard aligned with common prospective-LCA time horizons

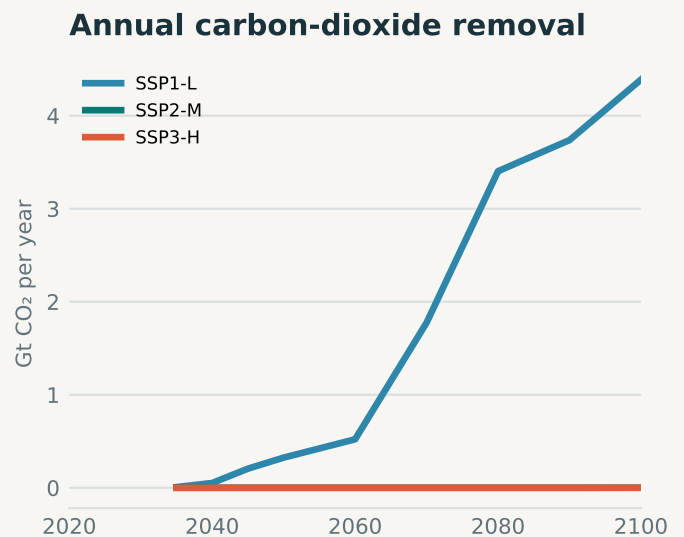
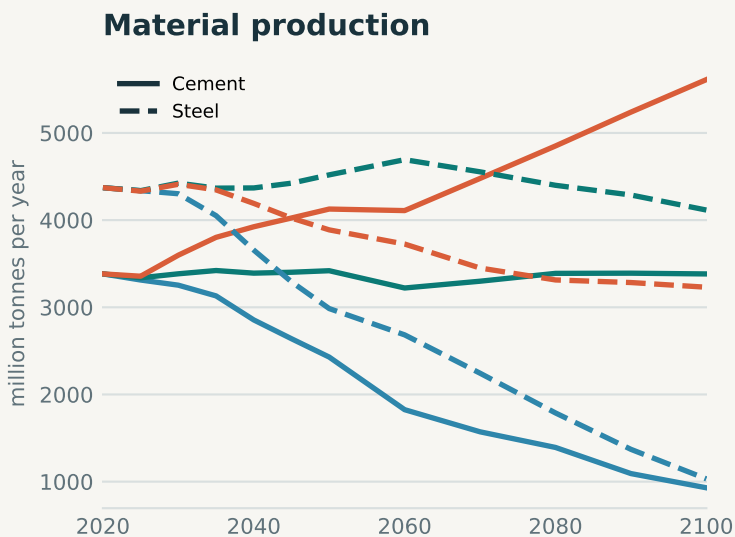
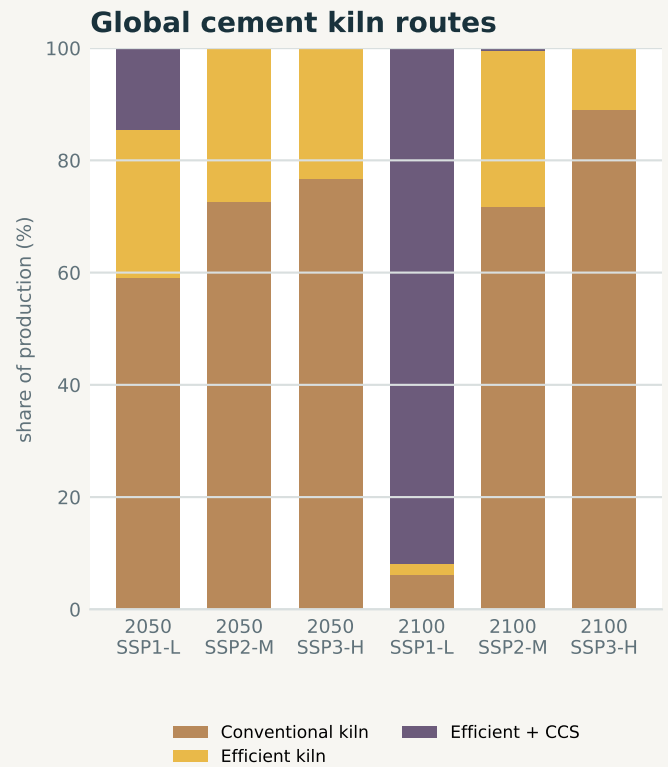
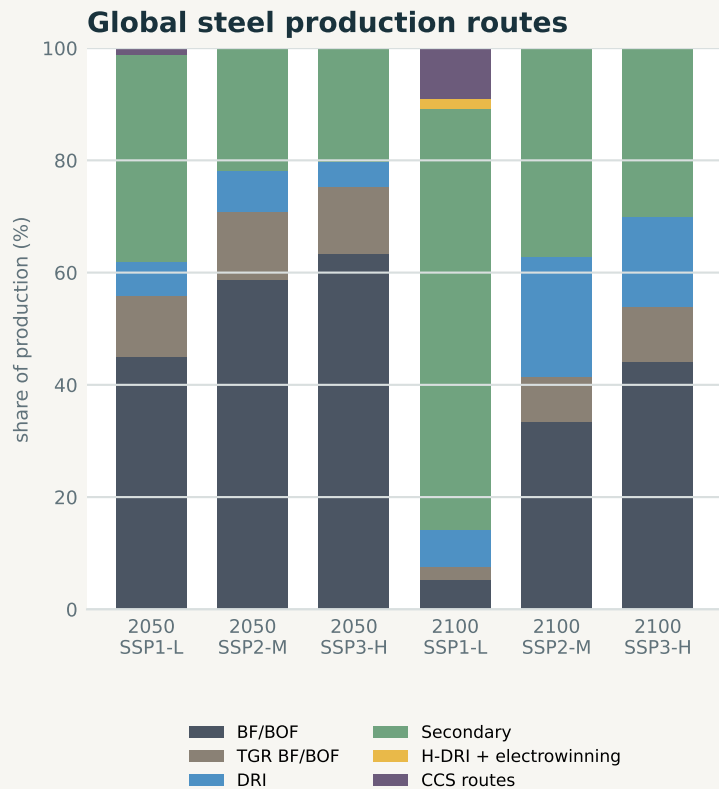
POPULATION		GDP PER PERSON		ANNUAL CO ₂		WARMING	
SSP1-L	9.2 bn	SSP1-L	\$31.2k	SSP1-L	11.9 Gt	SSP1-L	1.88°C
SSP2-M	9.6 bn	SSP2-M	\$27.5k	SSP2-M	34.7 Gt	SSP2-M	1.94°C
SSP3-H	10.1 bn	SSP3-H	\$22.8k	SSP3-H	48.9 Gt	SSP3-H	2.10°C

INDICATOR	UNIT	SSP1-L	SSP2-M	SSP3-H
Electricity generation	EJ/yr	214.6	223.3	192.5
Non-fossil electricity	% generation	84.9%	69.6%	67.6%
Electric and heat-pump heat	% heat supply	40.5%	31.4%	17.0%
Battery-electric car activity	% vehicle-km	51.9%	19.9%	2.0%
Efficient or CCS cement kilns	% production	40.9%	27.4%	23.3%
Secondary steel	% production	36.8%	21.8%	20.2%
Liquid fossil passenger energy	% final energy	59.0%	77.2%	88.0%

By 2050, SSP1-L is already a distinct technological world—not merely a lower-emissions version of SSP2-M. The largest differences are visible in electrification, industrial routes and fossil-fuel dependence.

Industry and carbon removal separate SSP1-L

Global production routes and removal deployment through 2100



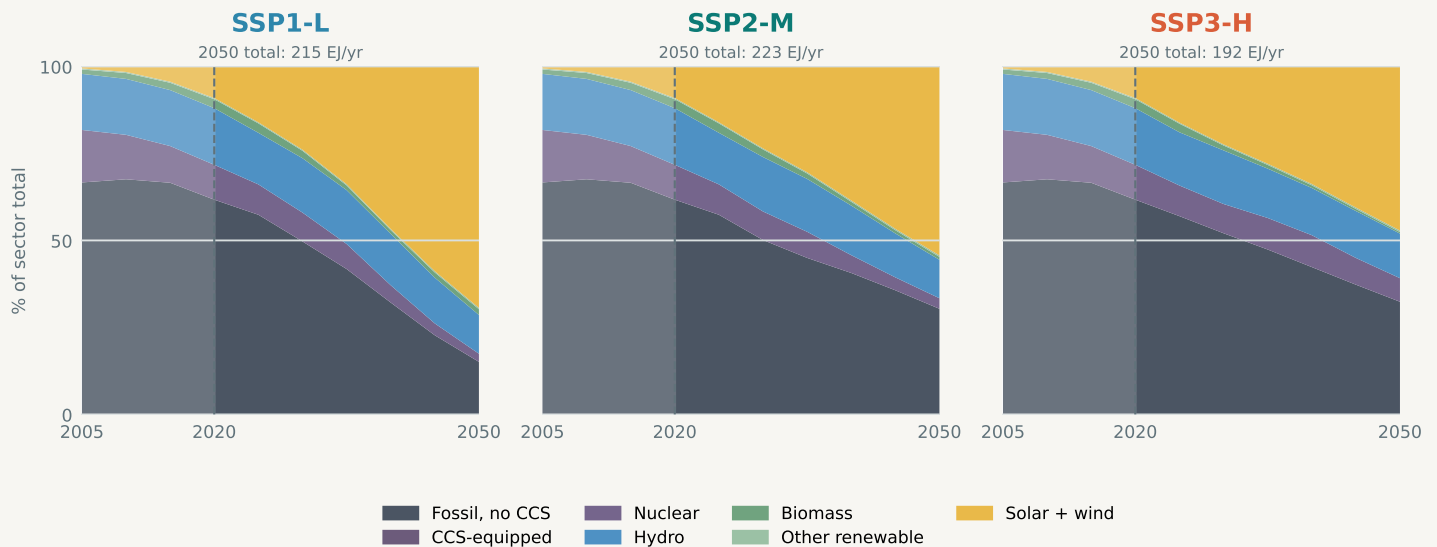
SSP1-L combines lower material demand with faster adoption of secondary steel, efficient kilns and CCS-equipped routes. CCS prevents some facility emissions from reaching the atmosphere; carbon-dioxide removal (CDR) removes CO₂ already in the atmosphere. They are not interchangeable, and only CDR can directly produce net-negative emissions.

Electricity: rapid divergence after 2020

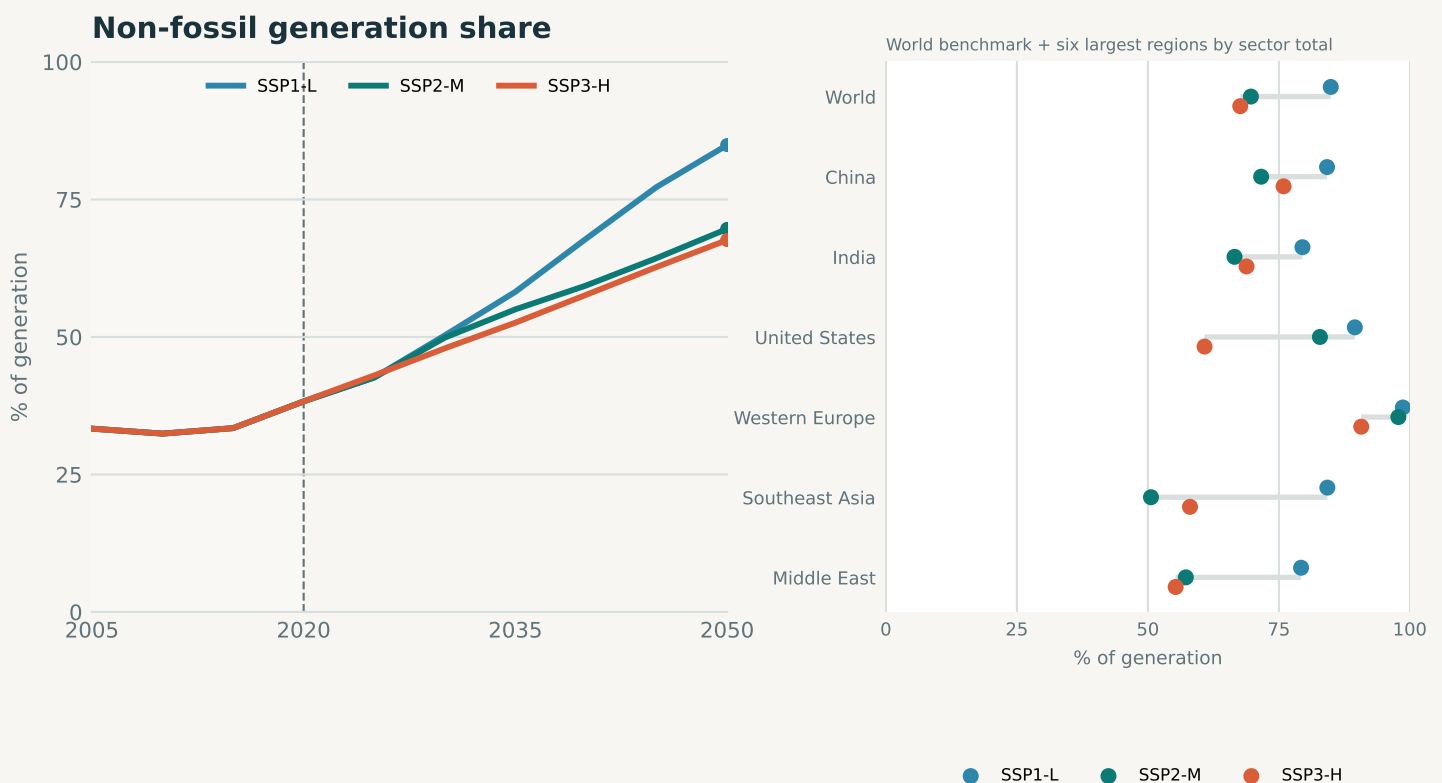
Technology mix, generation shift and regional variation

GLOBAL TECHNOLOGY MIX, 2005-2050 • SAME 0-100% SCALE

shaded = 2005-2020



Regional non-fossil share in 2050



By 2050, non-fossil electricity reaches about 85% in SSP1-L, compared with roughly 70% in SSP2-M and 68% in SSP3-H. SSP1-L also activates CCS-equipped power technologies that are absent or negligible in the other pathways.

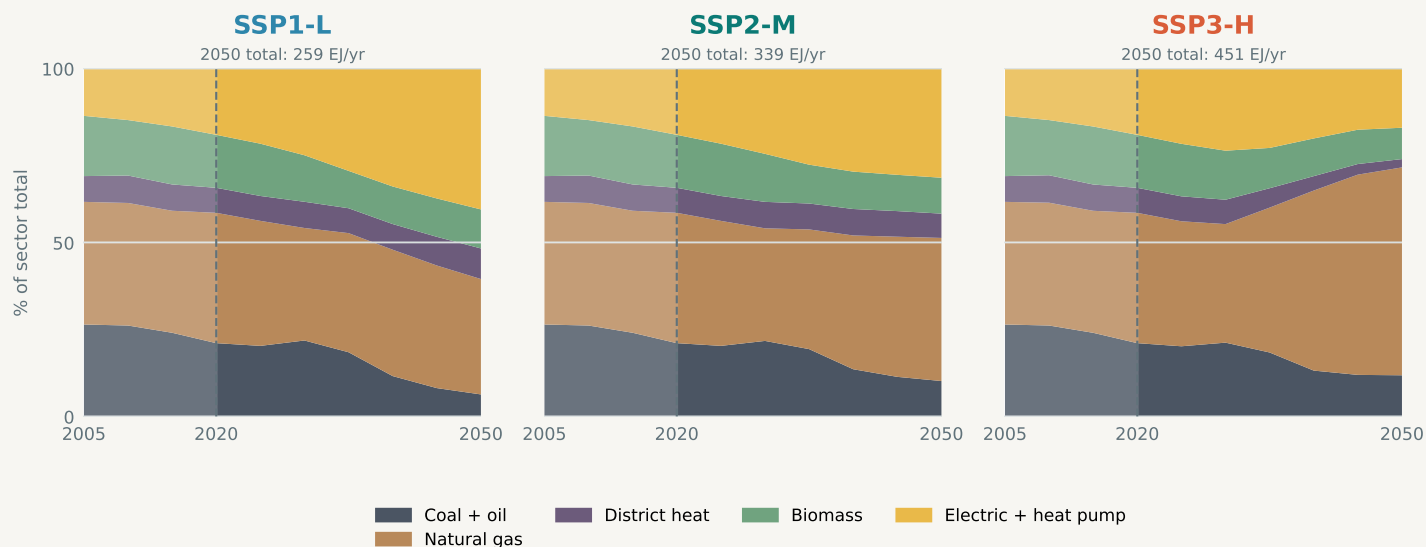
Technical power-plant efficiency series are not present in this CSV; technology shares are shown instead.

Heat: electrification competes with fossil boilers

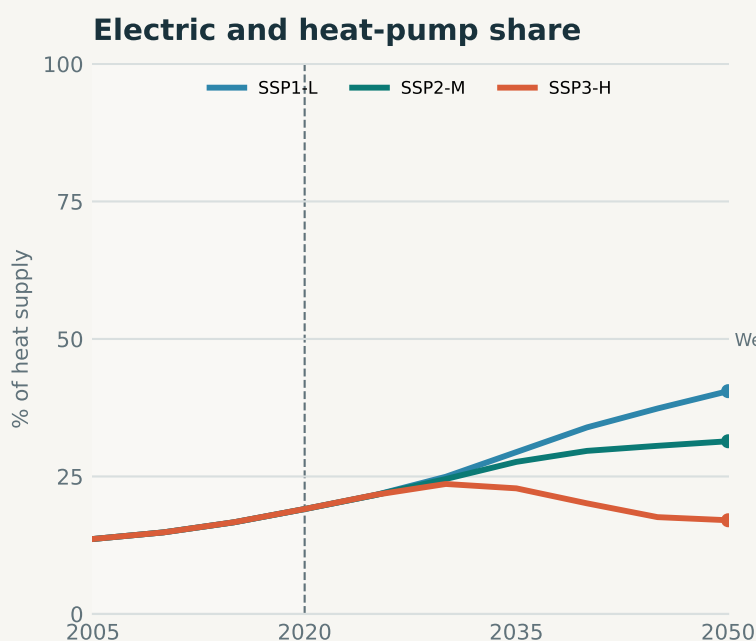
Building and industrial heat supply from 2005 to 2050

GLOBAL TECHNOLOGY MIX, 2005-2050 • SAME 0-100% SCALE

shaded = 2005-2020



Regional fossil share in 2050



World benchmark + six largest regions by sector total



SSP1-L combines lower heat demand with faster adoption of electric heating and heat pumps. By 2050, fossil sources provide about 40% of heat in SSP1-L, 51% in SSP2-M and 72% in SSP3-H; regional differences remain wider than the global averages suggest.

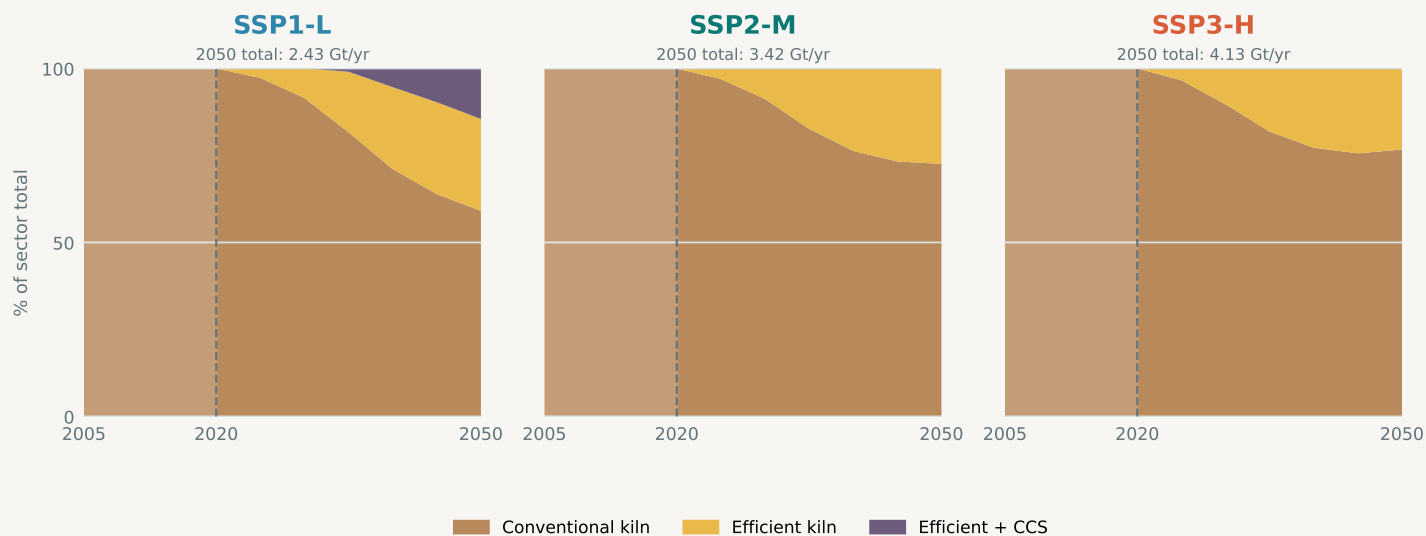
Technical boiler and heat-pump efficiency series are not present in this CSV; technology shares are shown instead.

Cement: efficient kilns and CCS scale up

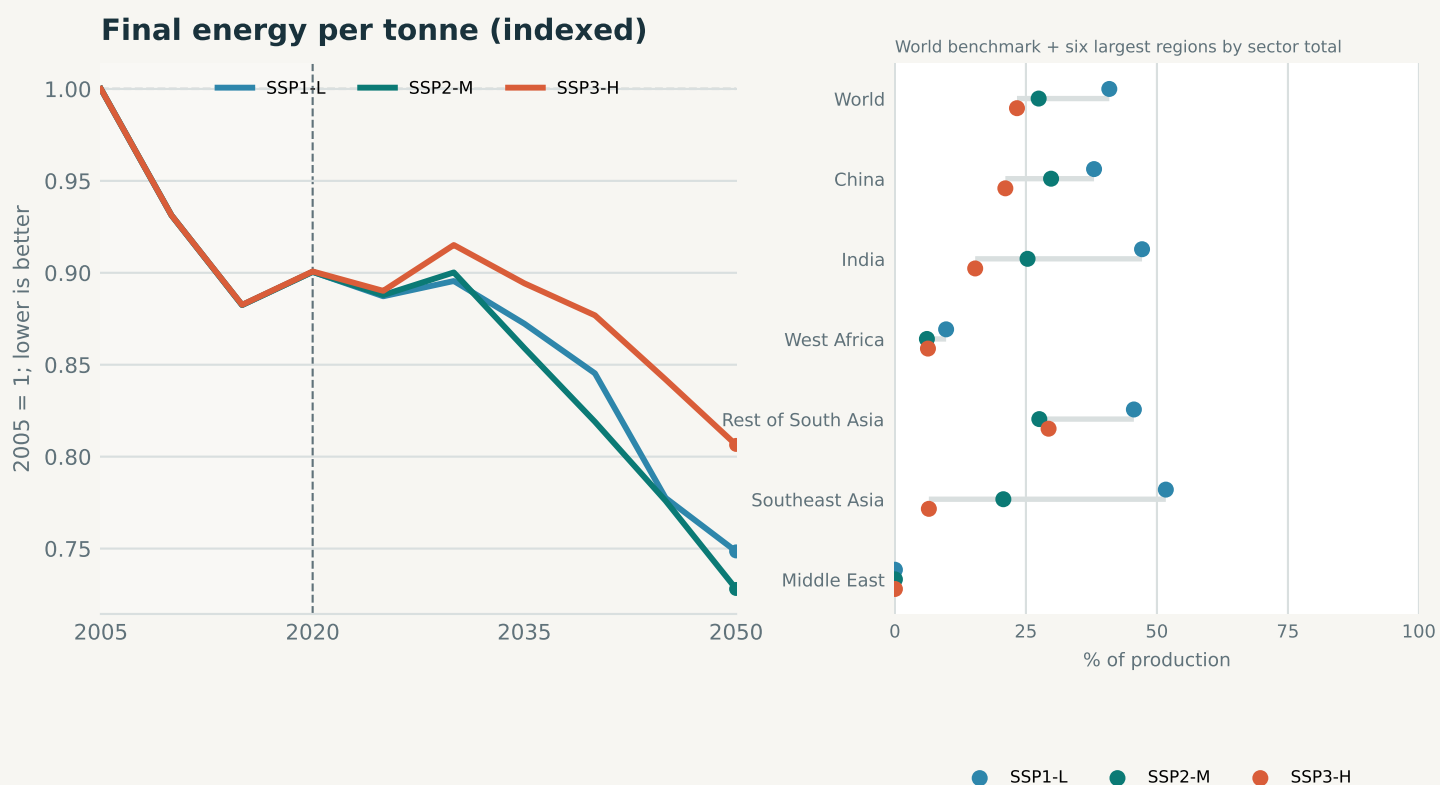
Production routes, energy intensity and regional adoption

GLOBAL TECHNOLOGY MIX, 2005-2050 • SAME 0-100% SCALE

shaded = 2005-2020



Efficient/CCS kiln share in 2050



SSP1-L produces substantially less cement and shifts about 41% of 2050 production to efficient or CCS-equipped kilns; explicit CCS supplies roughly 15%. SSP2-M and SSP3-H remain dominated by conventional kilns through mid-century.

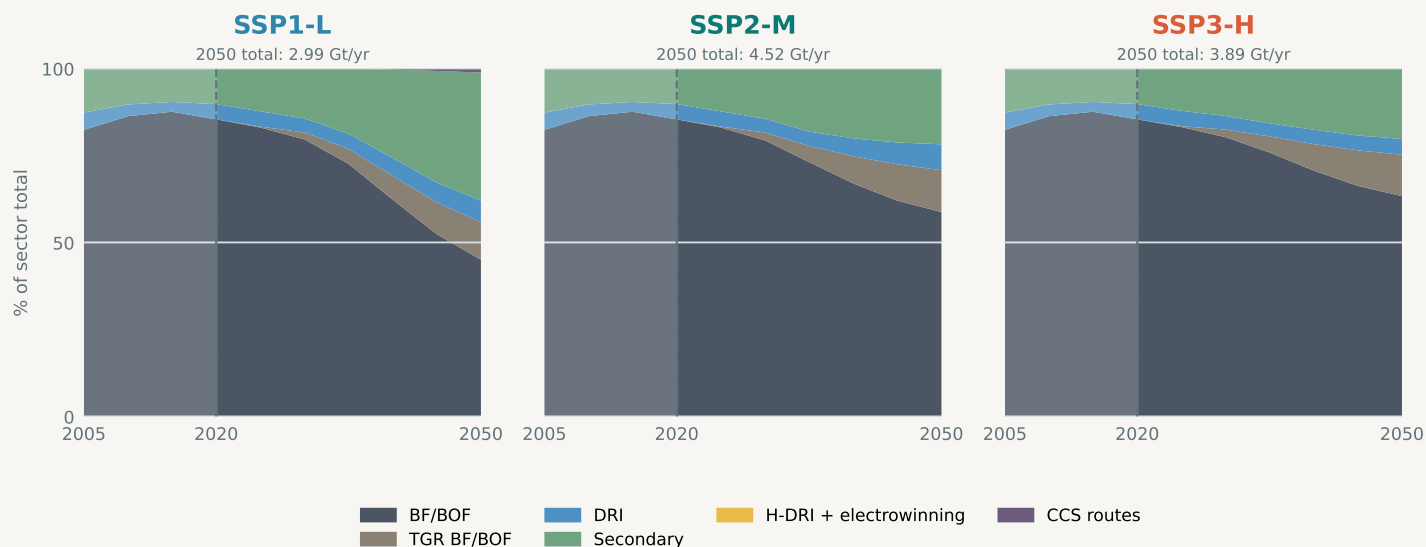
Derived sector average: lower energy per tonne does not guarantee lower emissions; kiln route, fuel mix and CCS energy needs also matter.

Steel: recycling leads, advanced routes emerge

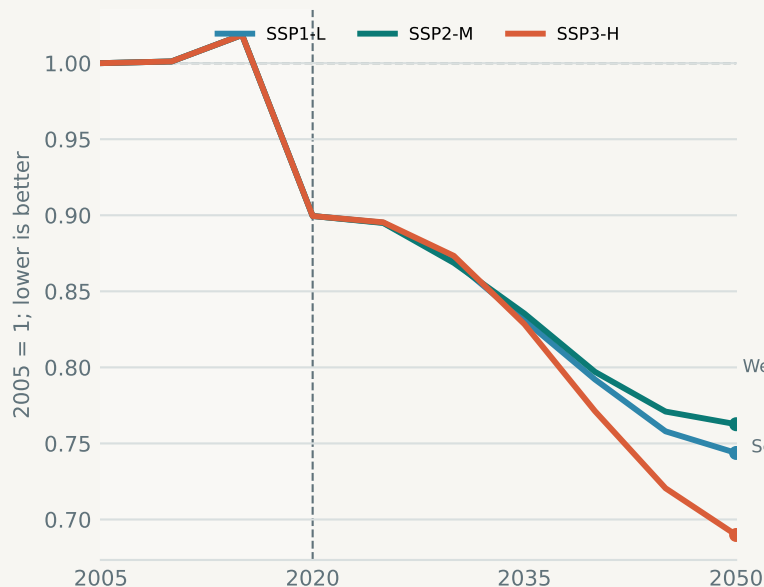
Production routes, energy intensity and regional adoption

GLOBAL TECHNOLOGY MIX, 2005-2050 • SAME 0-100% SCALE

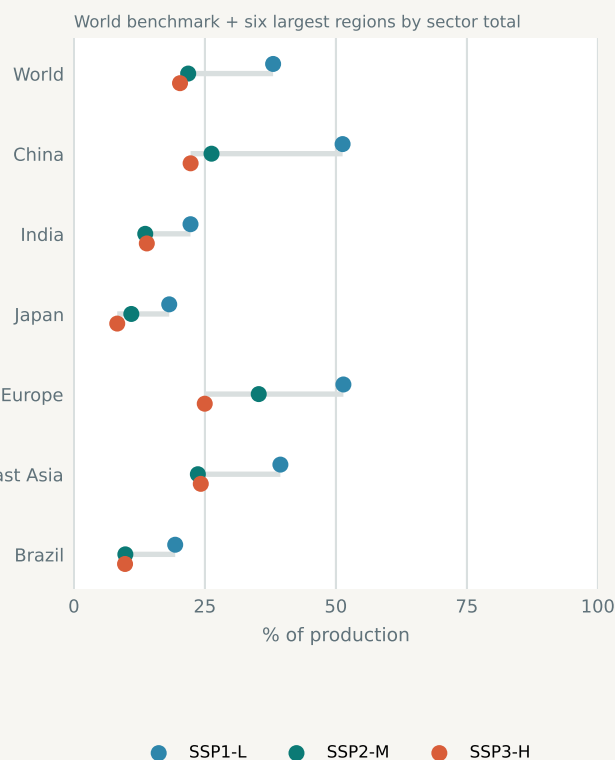
shaded = 2005-2020



Final energy per tonne (indexed)



Low-carbon route share in 2050



SSP1-L reaches about 37% secondary steel by 2050 and begins deploying H-DRI, electrowinning and CCS routes. SSP2-M and SSP3-H retain larger blast-furnace shares. DRI alone is kept separate because its climate performance depends on the reductant and electricity source.

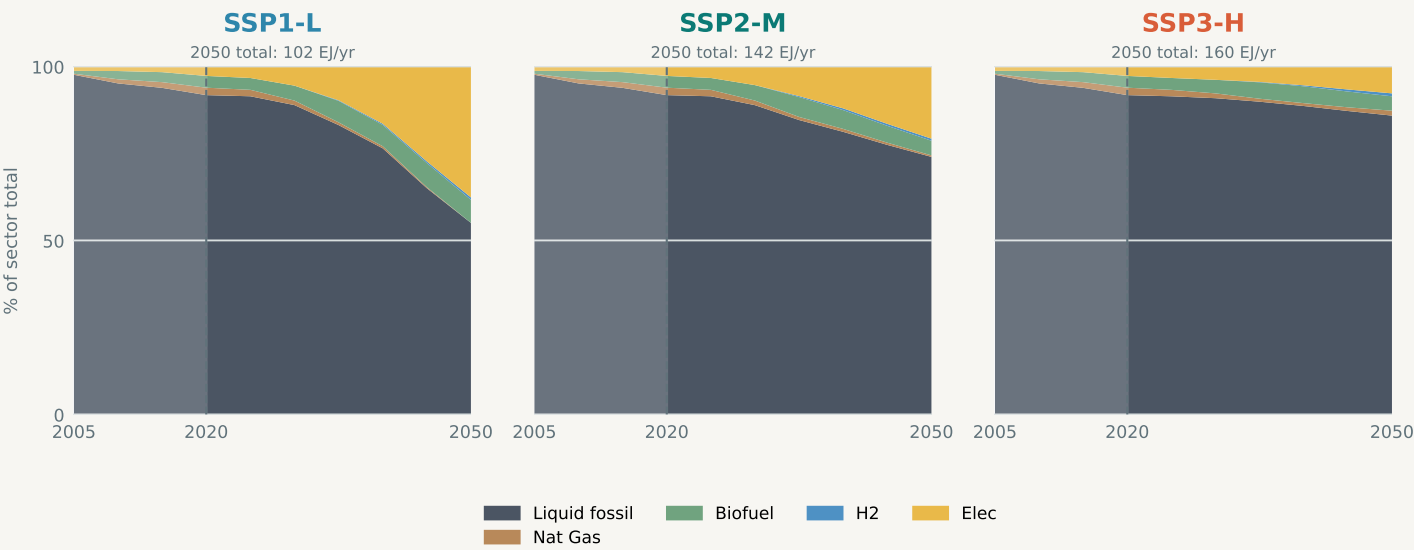
Derived sector average: lower energy per tonne does not guarantee lower emissions; route, reductant, electricity and CCS energy needs also matter.

Transport: electrification moves at three speeds

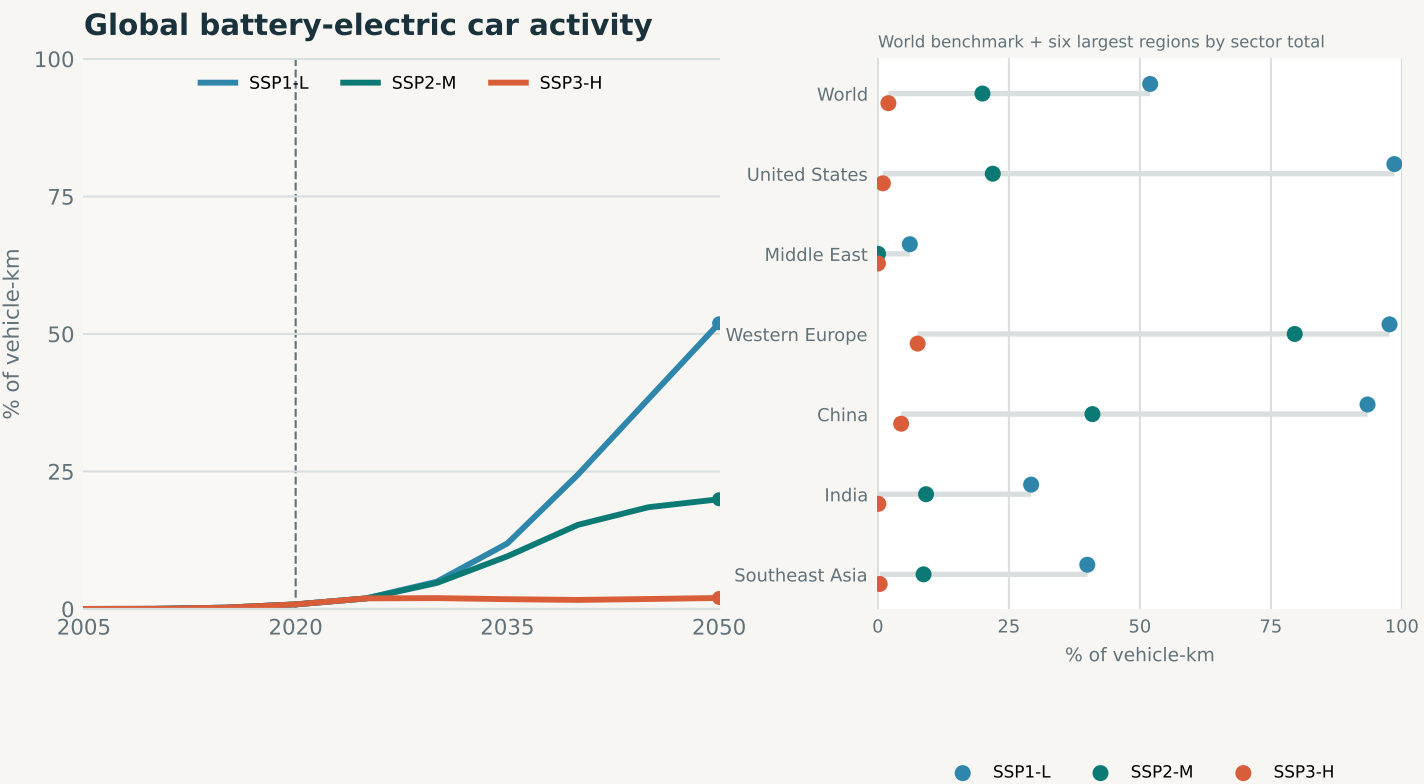
Energy mix, battery-electric activity and regional uptake

GLOBAL TECHNOLOGY MIX, 2005-2050 • SAME 0-100% SCALE

shaded = 2005-2020



Battery-electric activity in 2050



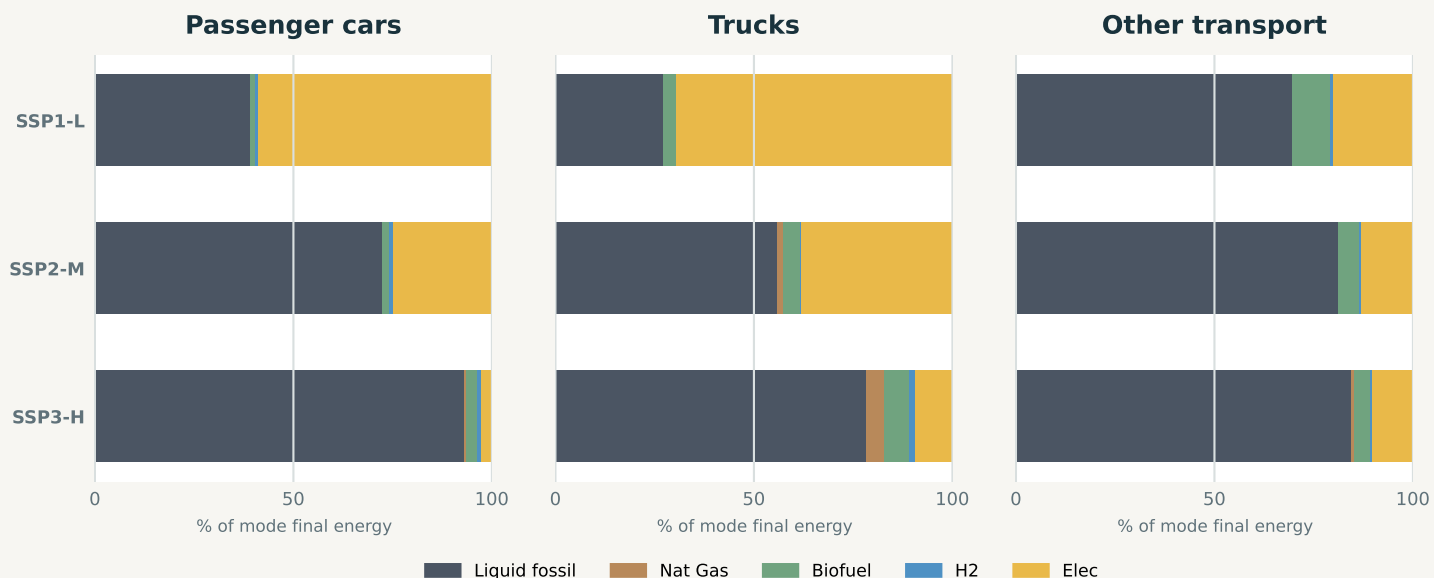
By 2050, electricity supplies 38% of total passenger and freight final energy in SSP1-L, while battery-electric cars provide 52% of passenger-car activity. The measures differ because final energy includes freight and electric cars use less energy per kilometre. SSP2-M and SSP3-H electrify much more slowly.

Vehicle efficiency series are not present in this CSV; activity share is not a fleet-stock, sales or test-cycle-efficiency measure.

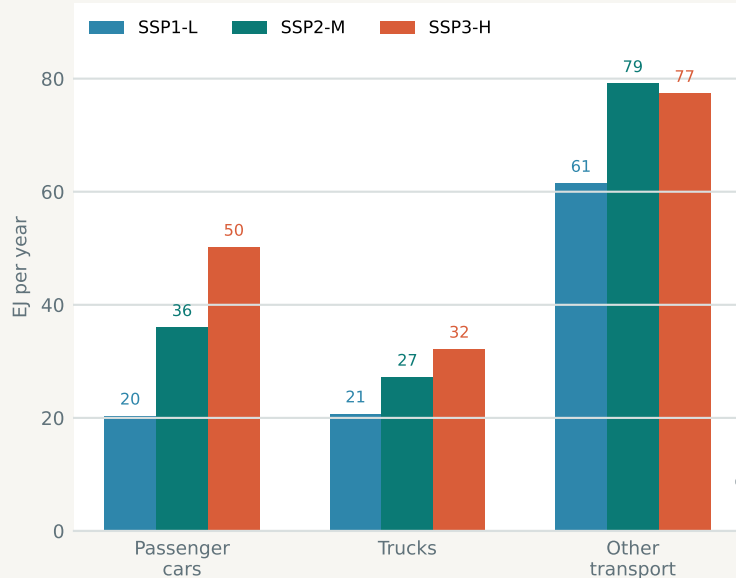
Cars, trucks and other transport follow different routes

Final-energy composition and demand in 2050 across the three IMAGE pathways

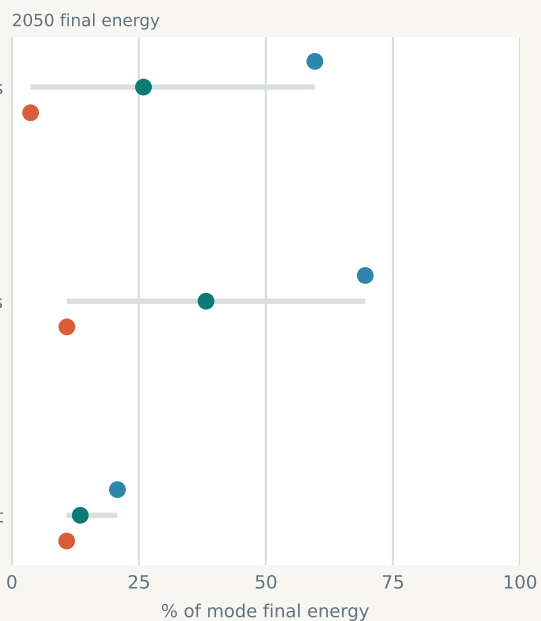
2050 FINAL-ENERGY MIX • SAME 0-100% SCALE



Absolute final-energy demand in 2050



Electricity + hydrogen share



In SSP1-L, electricity and hydrogen supply about 60% of passenger-car final energy and 70% of truck energy by 2050, but only 21% of other transport. Aviation, shipping, rail and buses together remain the largest energy block and still obtain about 70% of their energy from liquid fossil fuels.

'Trucks' combines 18- and 40-tonne road freight; 'other transport' combines buses, rail, aviation and shipping. Shares are final energy—not activity, fleet stock or life-cycle impact.